Sergio Cordoba (MSc. Physics, MSc. Mathematical modeling)
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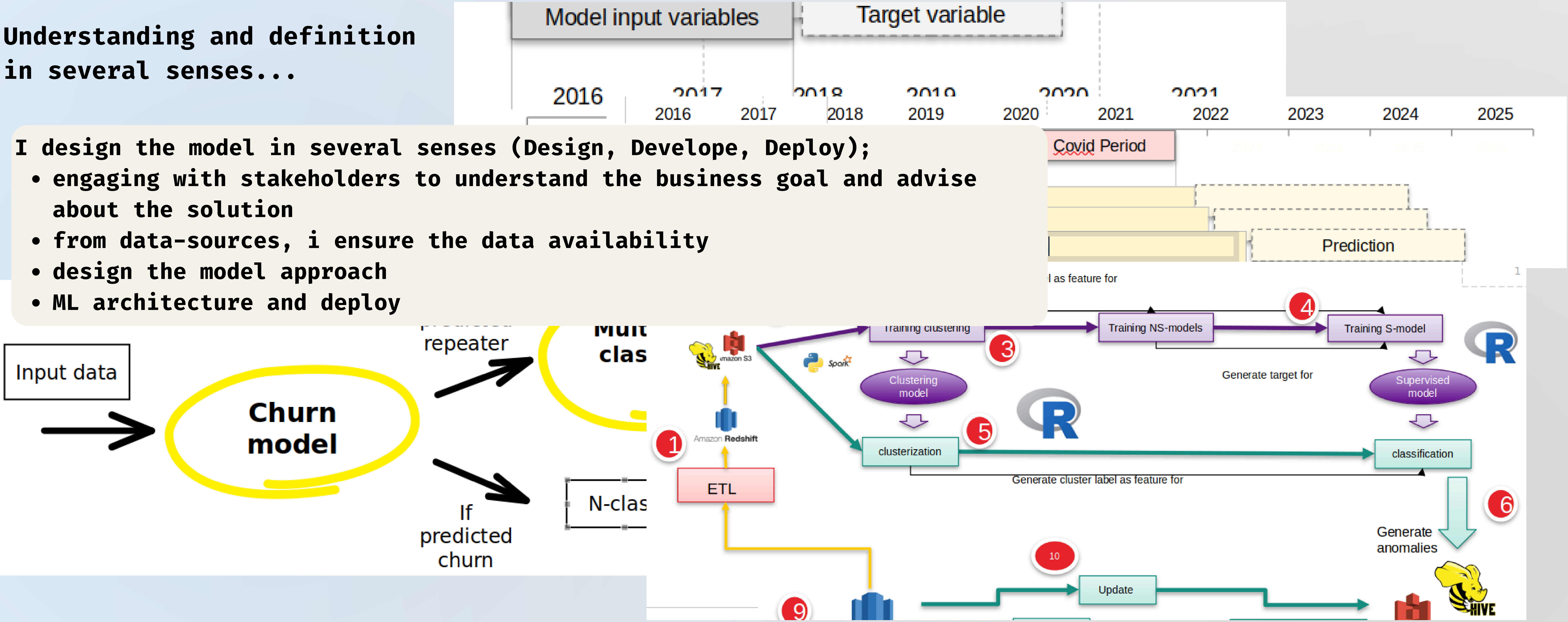
BUSINESS CASES EXAMPLES

data scientist | **+10 years experience** in differents sectors | **oriented to business** processes | **task & timings** management | **agile** methodologies
end-to-end data (science) pipelines design, building and operationalization | **monitoring** business-KPIs and model performance

Understanding and definition in several senses...

I design the model in several senses (Design, Develope, Deploy);

- engaging with stakeholders to understand the business goal and advise about the solution
- from data-sources, i ensure the data availability
- design the model approach
- ML architecture and deploy



** project steps based on CRISP-DM work-flow

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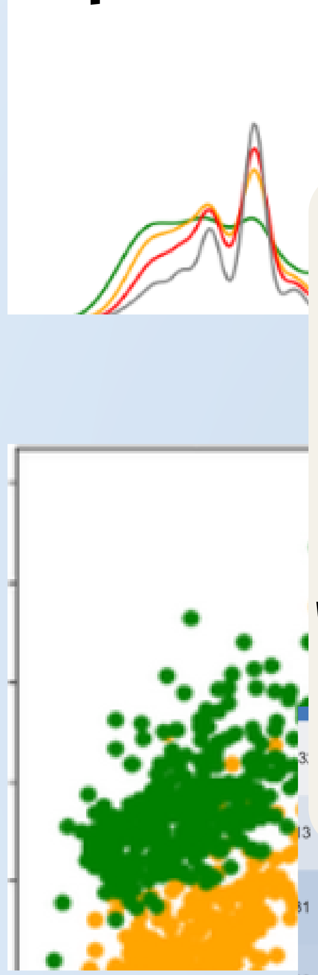
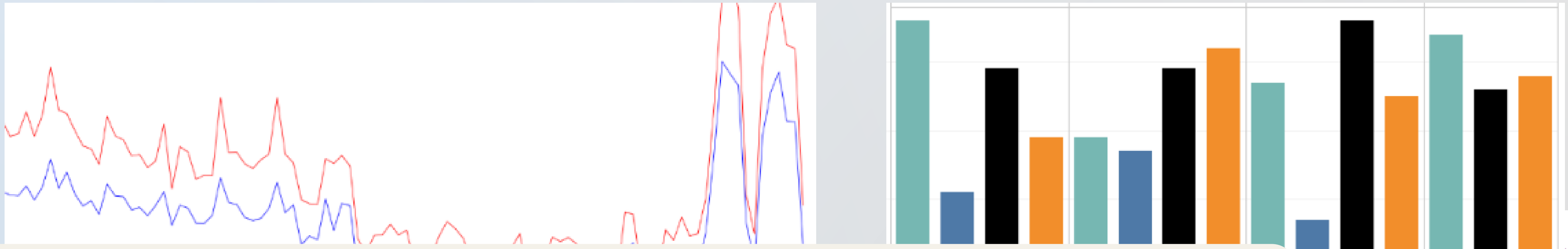
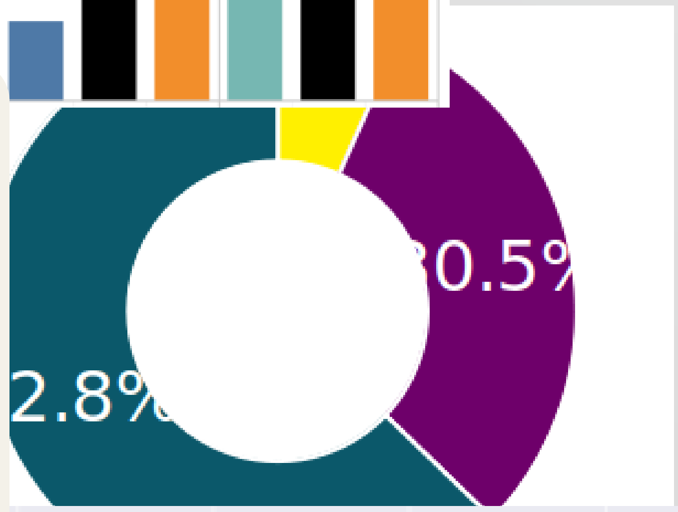
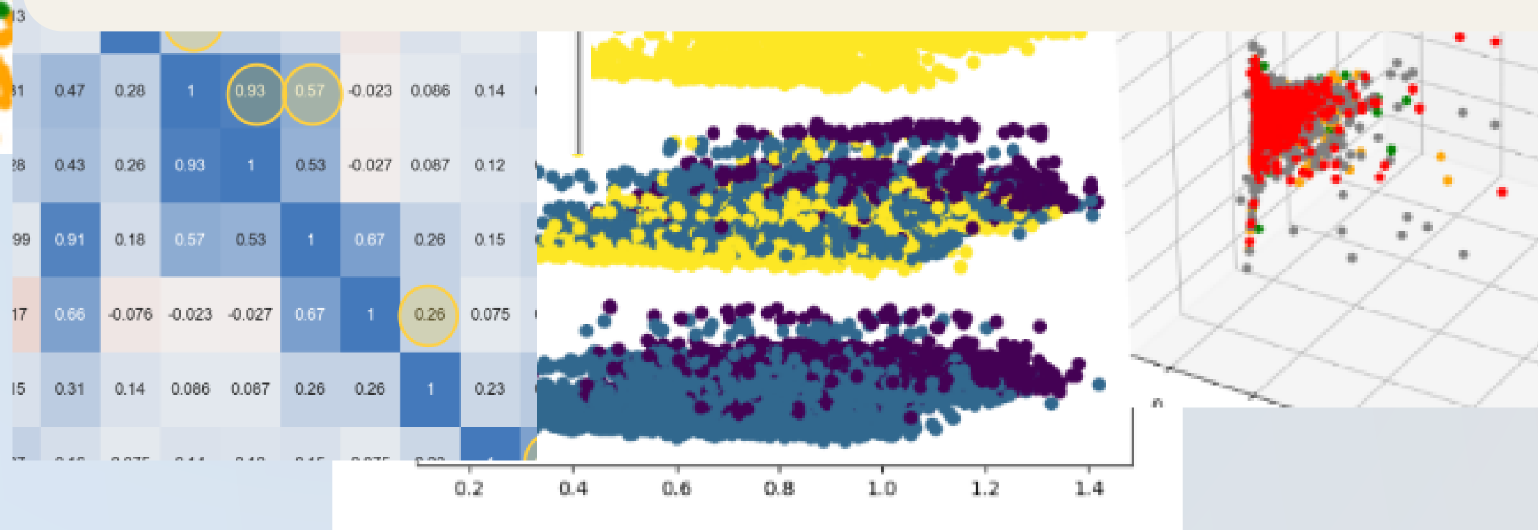
Data preparation and exploration...

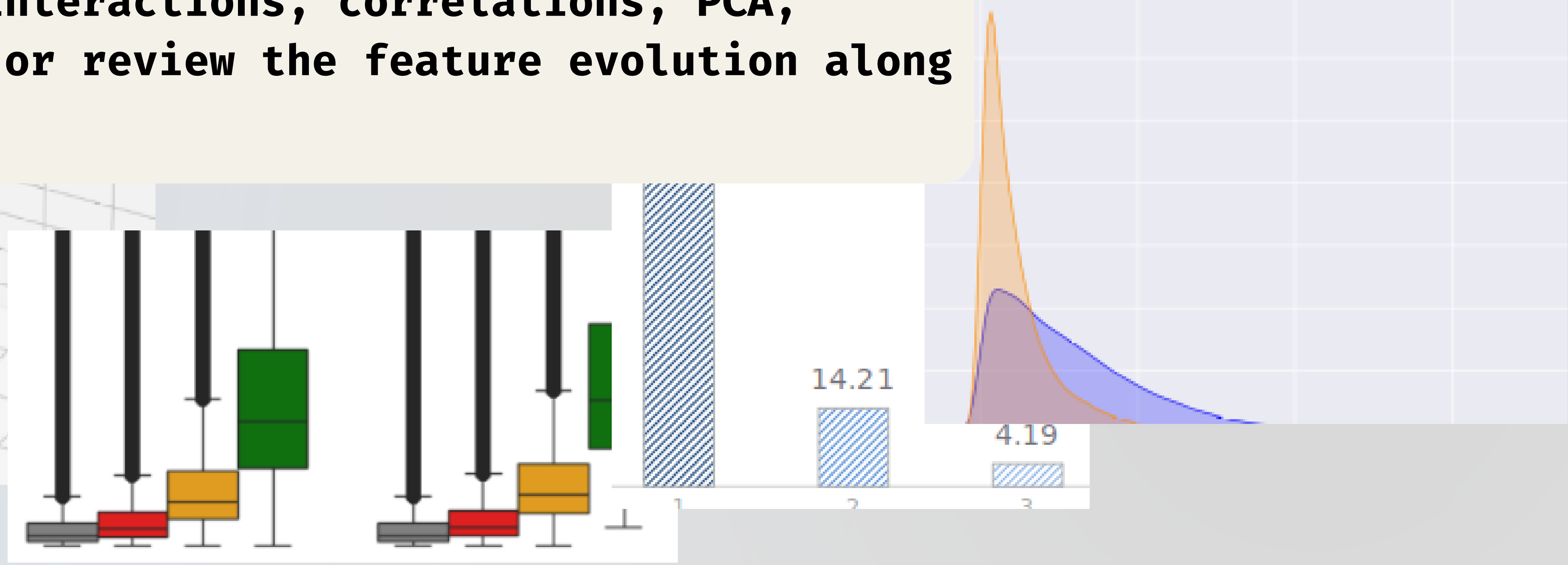






I perform presentations for business owners and/or technical teams (descriptives) Business insights are got and usefuls to data understanding and validation.
Distincts statistical or mathematical methods are applied in according with the specific case.
We can check univariate feature descriptives, interactions, correlations, PCA, target proportions, compare population samples or review the feature evolution along the time





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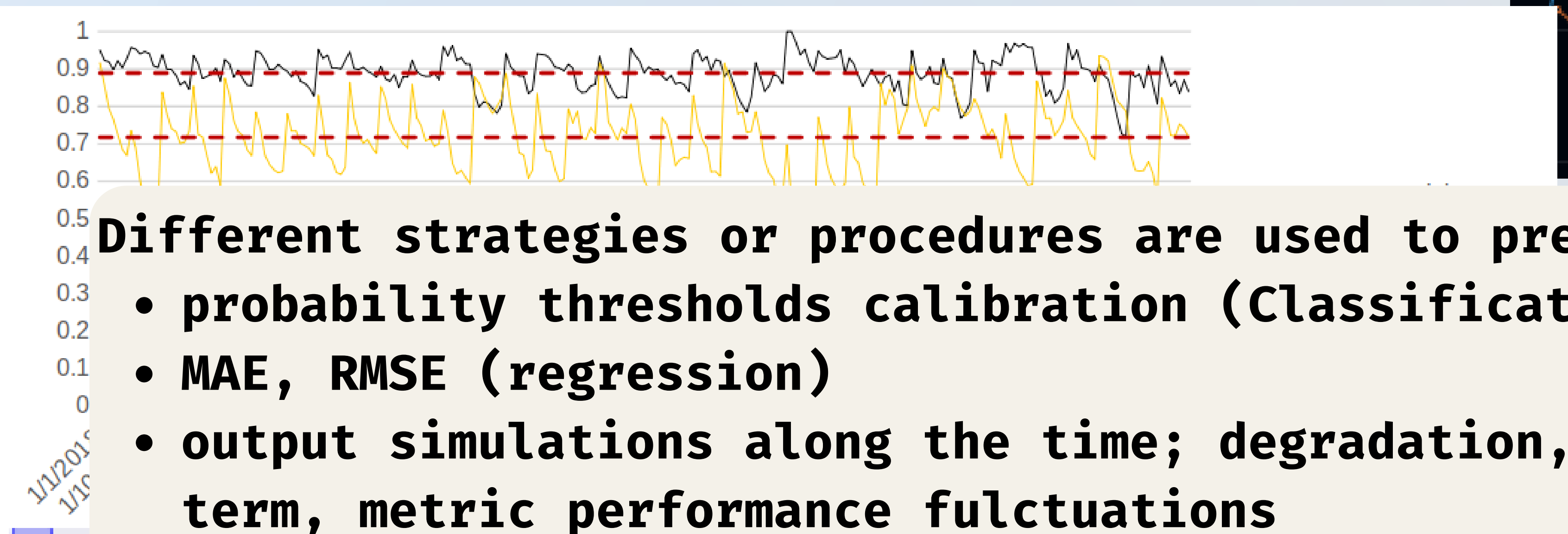
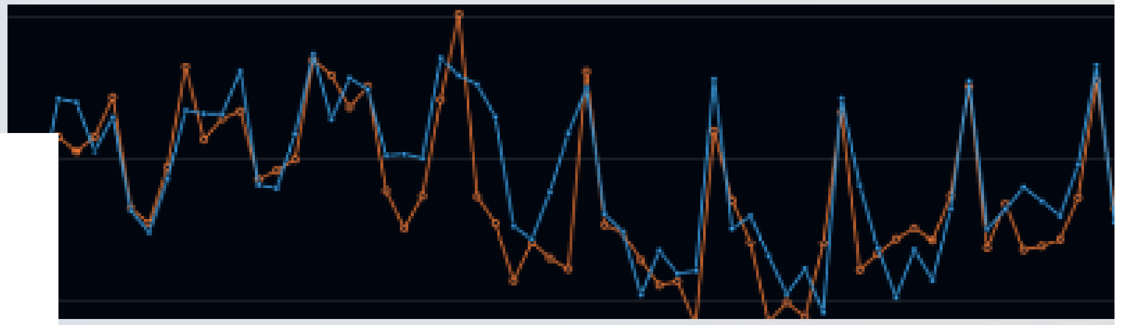
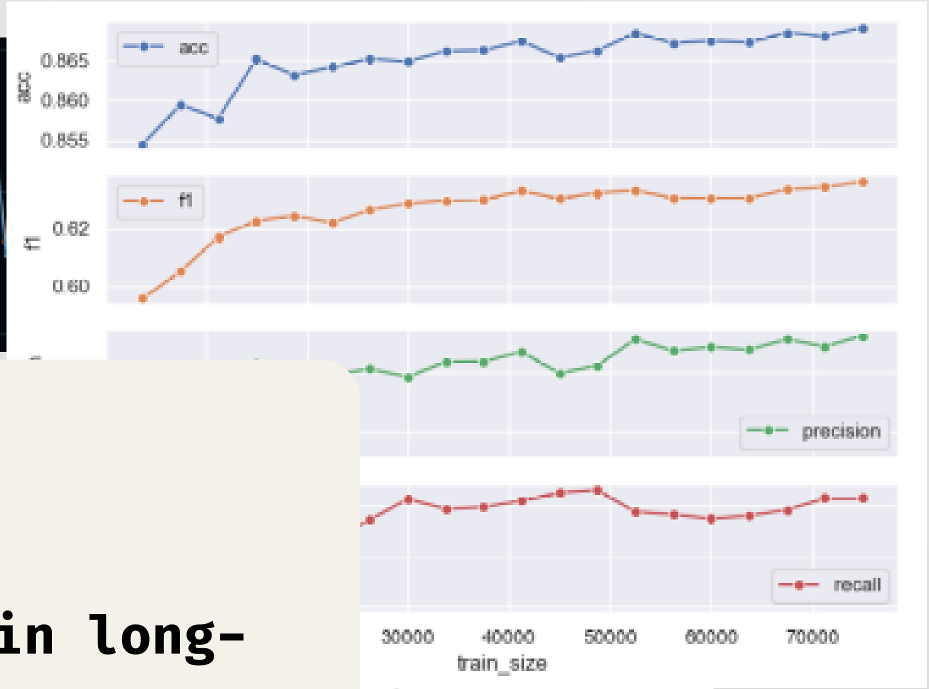
Modeling and evaluation

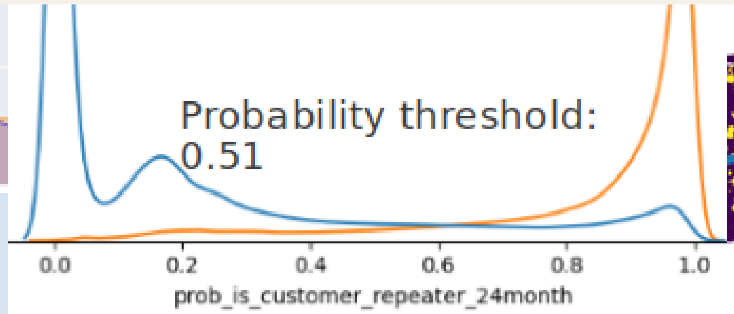


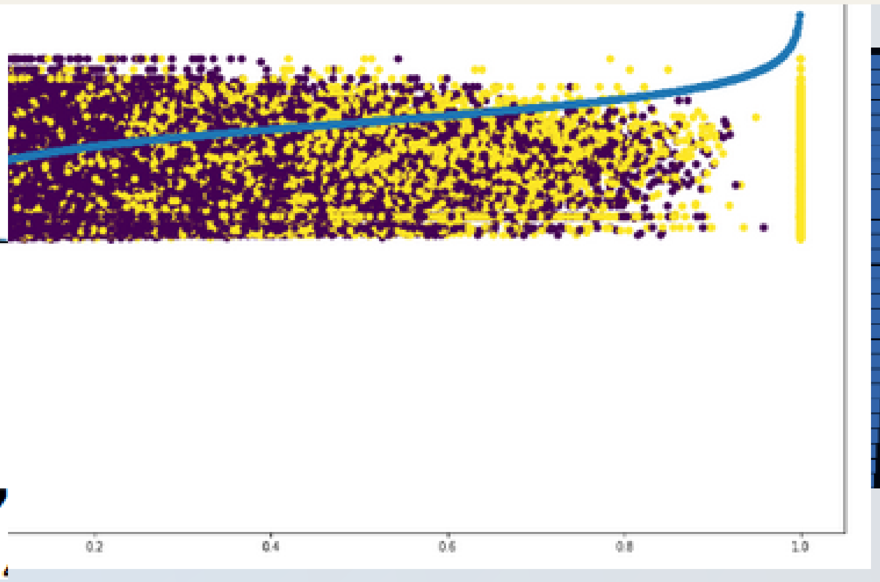


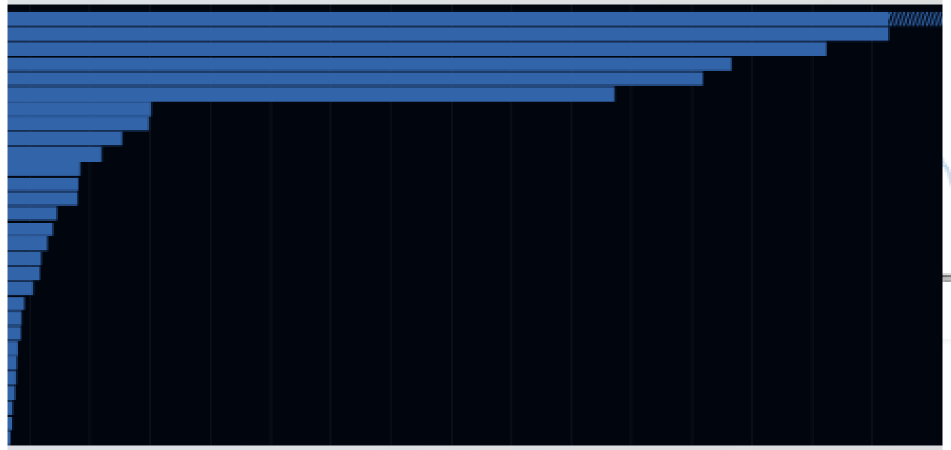


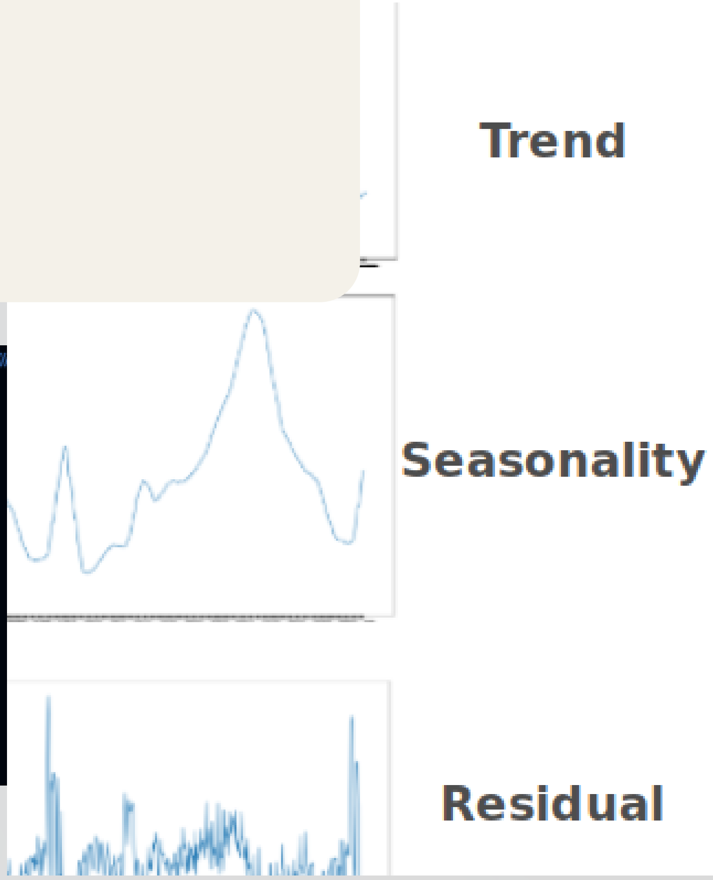
Different strategies or procedures are used to prepare and tune the model,

- probability thresholds calibration (Classification)
- MAE, RMSE (regression)
- output simulations along the time; degradation, ckeck the productive behaviour in long-term, metric performance fulctuations
- hyperparameters tuning
- ptimal sample size, to reduce the computational use (cost) can be checked
- Feature importance and selection, also, to optimize the training data volume









Good reliability,

global accuracy 86%,
F1=0.83

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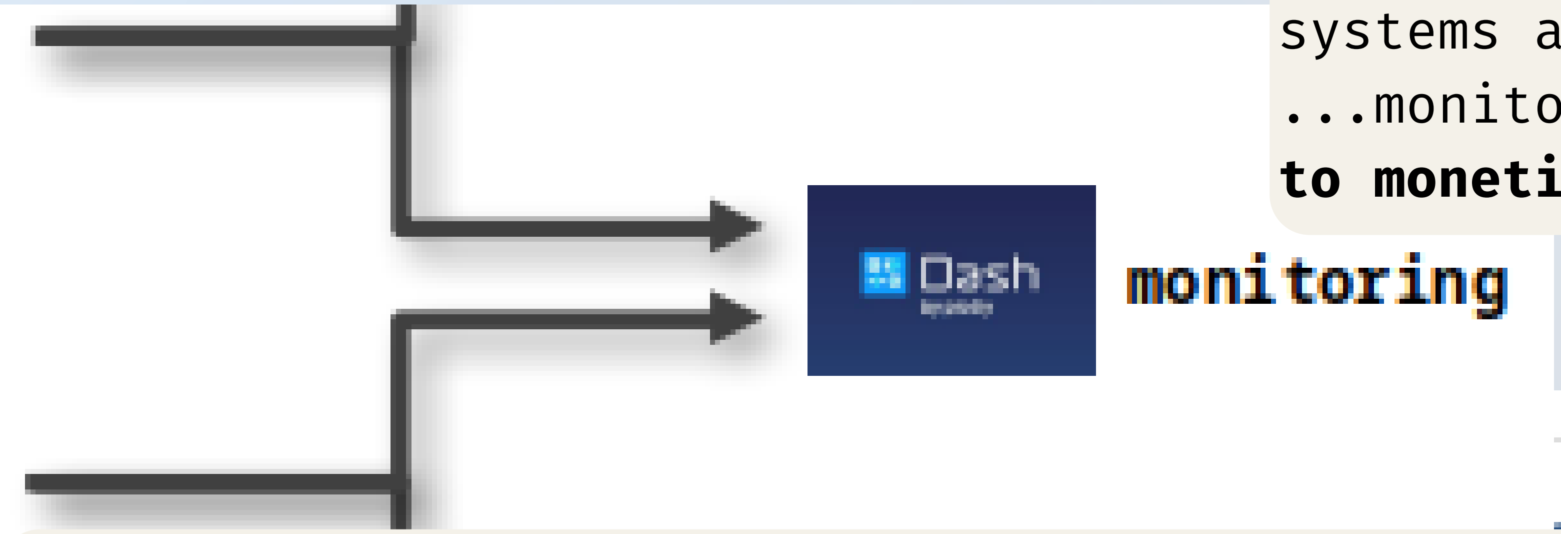
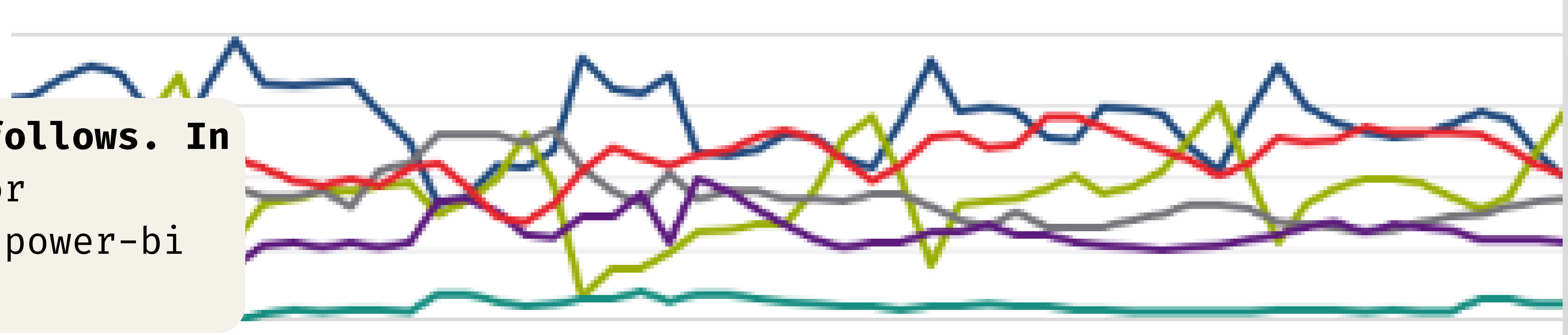
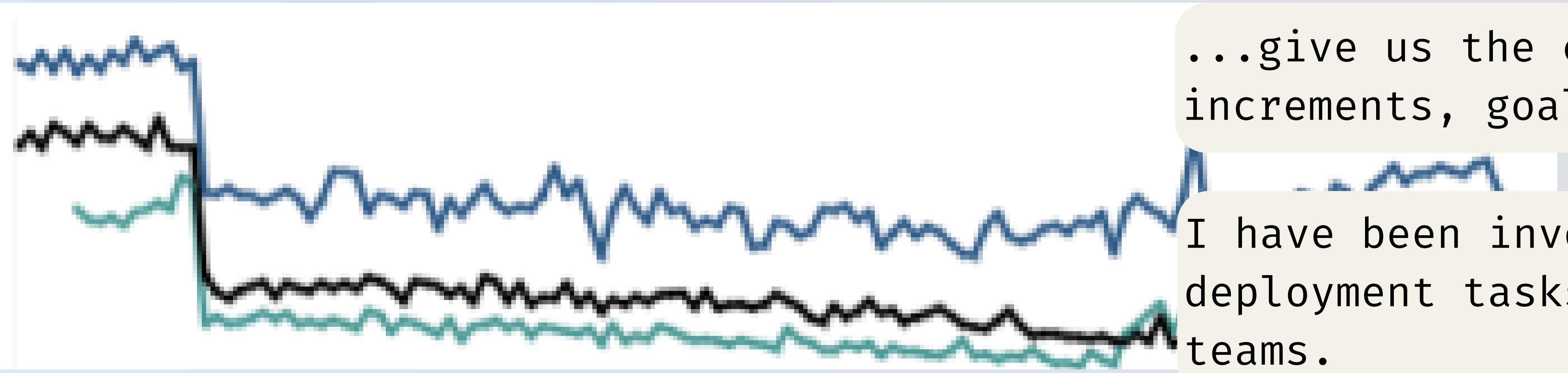
Deploy and monitoring

Productive models must be associated to control processes, in analytical, systems and business terms...
...monitoring along the time **becomes mandatory (and beautiful), to be able to monetize the project and track the model performance**





Different (technical or KPIS) **metrics could be follows. In this context, visualization tools can be used** for developers (i.e. grafana) and stakeholder (i.e. power-bi or tableau)...



...give us the capability to detect cycles, drastic changes, alerts, increments, goals...

I have been involved around projects standarization, visualization deployment tasks with IT and MLOps departments. Cross-functional teams.